Sameer Shashikant Rajendra

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EDUCATION

Stevens Institute of Technology – Hoboken, NJ May 2026
MS in Applied Artificial Intelligence: GPA 3.60

Symbiosis Skills and Professional University – Pune, India May 2021
Bachelor of Technology in Mechatronics

TECHNICAL SKILLS

GPU / Parallel Computing: CUDA, NVML, CUTLASS, FlashAttention-2, cuDNN, Nsight Systems
Programming Languages: Python, C++
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace, TorchVision, OpenCV, Transformers, vLLM
Tools & Platforms: Docker, Git, Linux, QuestDB, Jupyter Notebook

WORK EXPERIENCE

Oneirix Labs Pvt. Ltd. – Pune, India, Associate Engineer **Sept 2021 – June 2024**

- **Real-Time CV System:** Engineered a multi-task real-time computer vision system deployed on live CCTV feeds for Pune Metro, using PyTorch and TorchVision for four simultaneous inference tasks: (1) crowd density estimation at ticket counters, (2) platform intrusion and track-jump detection via pose estimation, (3) metro property damage detection, and (4) unattended baggage detection all under strict per-frame latency constraints on continuous high-resolution video streams.
- **C++ Performance:** Developed communication algorithms using C++ to facilitate low-latency data exchange between controller units and scheduler algorithms.
- **Medical Imaging:** Designed a deep learning model to detect coronary artery diameter reduction, balancing diagnostic precision with inference overhead on resource-constrained deployment hardware.
- **Architecture:** Deployed containerized computer vision traffic-management applications via Docker; optimized container architecture for resource efficiency and horizontal scalability.

PROJECTS

LLM Inference Optimization **Jan 2026 – May 2026**

- **Architected a custom Fused Grouped-Query Attention (GQA) decode kernel** in C++/CUDA for Hopper (sm_90) architecture, designed specifically to break the LLM inference Memory Bandwidth Wall.
- **Implemented macro-sequence SRAM tiling** (64-token blocks) and cooperative thread-block memory fetches to maximize 128-byte HBM3e bus coalescing efficiency.
- **Eliminated 98% of block synchronization barriers** by replacing shared memory handoffs with high-velocity intra-warp register shuffles, achieving a **5.2x latency speedup** (3.13ms to 0.60ms) over naive baselines.
- **Profiled distributed PEFT (FSDP + LoRA) execution timelines** on a dual-H200 cluster using NVIDIA Nsight Systems (nsys), isolating compute-communication overlap across the 900 GB/s NVLink fabric.

Adaptive Evolutionary GPU Optimization Framework (AEGOF)- Project Report **Sept 2025 – Dec 2025**

- Architected an automated autotuning pipeline to jointly optimize 8 hardware/software "knobs" (Power Limits, Batch/Seq, Kernel Fusion) for **Llama-2-7B** on **NVIDIA H100** clusters.
- Developed a custom **NSGA-II** evolutionary core using the **DEAP** library to solve multi-objective trade-offs between throughput (tokens/s) and power consumption (Watts).
- Integrated **FlashAttention-2** and **CUTLASS** backends via a white-box benchmarking harness, achieving up to an **8.09× efficiency multiplier** over standard PyTorch defaults.
- Implemented robust systems-level error handling for **CUDA Out-of-Memory (OOM)** and **Subprocess timeouts**, enabling the evaluation of 238+ unique hardware configurations without manual intervention.

Financial-News-Sentiment-Analysis using GPT-Neo 125M- Project Report **Jan 2025 – May 2025**

- **Fine-tuned a GPT-Neo 125M model** as a sentiment classifier with Hugging Face and PyTorch, achieving **92% accuracy** and a **0.72 macro F1-score** on a test set of 36,000+ financial news articles.
- Developed a data pipeline to process articles from QuestDB, implementing a custom WeightedTrainer with a **weighted loss function** to correct for severe class imbalance.
- Packaged the trained transformer model and tokenizer into a streamlined inference function for **on-demand sentiment classification** and confidence scoring.

Detecting Diabetic Retinopathy using Computer Vision Project Report **Jan 2025 – May 2025**

- Developed a deep learning algorithm to detect Diabetic Retinopathy, achieving a peak validation accuracy of over 80.49% for rapid and reliable screening.
- Addressed the inherent class imbalance of the dataset by integrating sklearn's balanced class weights directly into the nn.CrossEntropyLoss function.
- The final evaluation highlights strong performance on the majority class ("No DR" F1-score: 0.91) and detailing the performance trade-offs on the imbalanced minority classes.

Awards & Recognition

HACK MIT-WPU 2024, CCTV Video Analytics Hackathon **May 2024**

Maha Metro × Ministry of Education, Maharashtra. MIT World Peace University, Pune- Team Oneirix Labs. Built a real-time multi-task CCTV analytics system for Pune Metro covering crowd estimation, platform intrusion detection, property damage detection, and unattended baggage detection recognized for innovation in sustainable urban transit infrastructure.